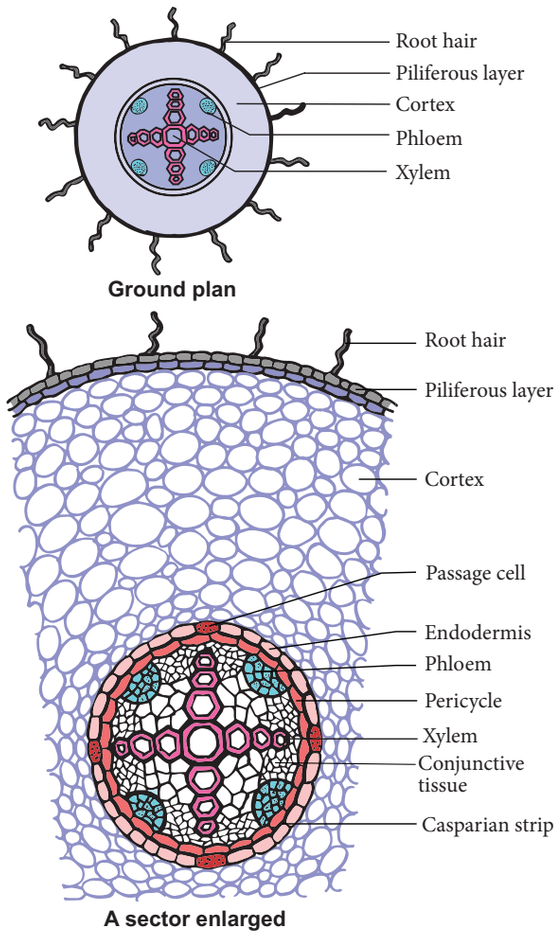


Figure 12.2 Transverse section of Dicot root

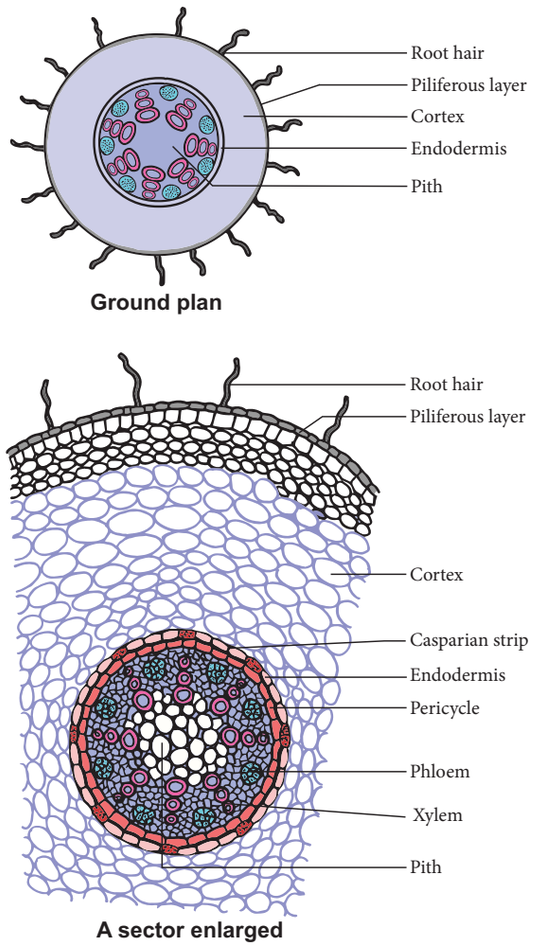


Figure 12.3 Transverse section of Monocot root

ii. Cortex: It is multilayered large zone, composed of parenchymatous cells with intercellular spaces. It stores water and food material.

iii. Endodermis: It is the innermost layer of cortex with characteristic casparian strips and passage cells. **Casparian strips** are band like thickening made of **suberin**.

iv. Stele: All the tissues inner to endodermis constitute stele. It includes pericycle, vascular tissues and pith.

a) Pericycle: It is a single layer of thin walled cells. The lateral roots originate from this layer.

b) Vascular tissues: It consists of many patches of xylem and phloem arranged radially. The xylem is exarch and polyarch. The conjunctive tissue is made up of sclerenchyma.

c) Pith: It is present at the center. It is made up of parenchyma cells with intercellular spaces. It contains abundant amount of starch grains. It stores food.

12.5 Internal Structure of Dicot Stem (Sunflower)

The transverse section of a dicot stem reveals the following structures.

1. **Epidermis:** It is the outermost layer. It is made up of single layer of parenchyma cells, its outer wall is covered with cuticle. It is protective in function.
2. **Cortex:**- It is divided into three regions:

(i) **Hypodermis:** It consists of 3 - 6 layers of collenchyma cells. It gives mechanical support.

10th_Science_Unit-12.indd 176 11-01-2020 15:36:11